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PROJECT TYPE: **DBMS PROJECT** STUDENT: NITHICK S J
PROJECT GUIDE: **Ms.Abishree.S AP/CSE**

PROJECT TITLE: **MEMBERPRO — AUTOMATED MEMBERSHIP MANAGEMENT AND BILLING SYSTEM**

Abstract

The Automated Membership Management and Billing System (MemberPro) is a modern, full-stack, cloud-database web application designed to simplify, automate, and streamline the operational governance of subscription- and membership-based organizations (such as fitness centers, clubs, libraries, co-working spaces, and professional bodies). The system establishes a centralized platform for managing subscriber user accounts, subscription tiers and plans, membership lifecycles, offline/online payment audits, renewal tracking, and automated expiration monitoring. It aims to eliminate error-prone manual spreadsheets and paper receipts, enhance administrative efficiency, and deliver a frictionless self-service portal connected to a secure relational database (PostgreSQL on Supabase).

The application is structured into two dedicated, role-protected portals: the Member Self-Service Portal and the Administrative Management Portal. Members can register personal credentials, browse available subscription tiers with active pricing, self-enroll into membership plans, view days remaining until expiration, monitor complete transactional billing logs, submit renewal requests, or cancel subscriptions on demand. The administrative portal equips authorized staff with real-time operational analytics (total members, active/expired counts, memberships expiring within 5 days, monthly revenue, and overall gross turnover), a dedicated member registry with comprehensive status management (activate, suspend, cancel), an offline payment recording desk, and renewal approval workflows.

The system is engineered using a modern Single Page Application (SPA) architecture comprising React, Vite, Tailwind CSS, JavaScript (ES6+), and PostgreSQL (Supabase). Supabase handles database hosting, user authentication, and granular data security via Row-Level Security (RLS) and Database Stored Procedures (PL/pgSQL Functions), while React provides an intuitive, high-performance, responsive user interface. The project demonstrates a secure, production-grade approach to database-driven subscription management, automated constraint validation, and real-time operational analytics.

Objectives

- To develop a centralized, database-driven web application for managing membership lifecycles, subscription tiers, and payment auditing.
- To provide members with an interactive self-service dashboard to browse plans, track validity dates, view countdowns to expiration, and review payment receipts.
- To enable member self-enrollment, plan upgrades/downgrades, and membership cancellations via secure database stored procedures.
- To provide administrators with real-time business intelligence metrics, including monthly revenue, total turnover, active subscriptions, and churn rates.
- To implement proactive expiration tracking by calculating and alerting administrators of memberships expiring within 5 days.

- To provide an offline payment recording module supporting various payment modes (Cash, UPI / Bank Transfer, Card Swipe) with automated membership validity extension.
- To enforce robust Role-Based Access Control (RBAC) separating administrative privileges from regular member capabilities.
- To ensure enterprise-grade data security using PostgreSQL Row-Level Security (RLS) policies and Parameterized Database RPCs.
- To eliminate data redundancy and preserve relational integrity using strict foreign key constraints, unique indexes, and automated status triggers.
- To deliver a responsive, clean, light-themed user experience accessible across desktops, laptops, tablets, and smartphones.

Work Completed

- Designed and deployed the relational database schema in PostgreSQL/Supabase including tables for users, plans, memberships, payments, and renewal_requests.
- Established foreign key relationships with cascading integrity constraints (ON DELETE CASCADE) linking users and memberships to corresponding plans and transactions.
- Configured PostgreSQL Row-Level Security (RLS) policies guaranteeing that members can strictly query and modify only their own subscription records while admins retain full access.
- Implemented secure database stored procedures (update_my_plan, cancel_my_membership) running with SECURITY DEFINER privileges to safely execute multi-table transactions.
- Configured automated user synchronization between Supabase Auth (auth.users) and the application profile table (public.users).
- Developed a responsive frontend architecture using React, Vite, and Tailwind CSS styled with an uncluttered, high-legibility light theme.
- Implemented a client-side authentication context (AuthContext.jsx) managing persistent user sessions, JWT token lifecycle, and role detection.
- Built protected client routing (ProtectedRoute.jsx) to enforce role-based URL guards preventing unauthorized portal access.
- Created the Member Portal Dashboard (/member):
  - Profile card showing user details with an inline modal for profile updates.
  - Interactive subscription tier catalog showing plan validity, pricing, and instantaneous plan activation.
  - Active membership status card displaying plan name, start date, expiry date, and real-time validity countdown.
  - Plan upgrade, downgrade, and self-cancellation workflows.
  - Complete, chronological payment audit history table with amount, payment mode, and transaction remarks.
- Created the Admin Analytics Dashboard (/admin):
  - Real-time metric cards computing Total Members, Active Members, Expired Accounts, Expiring Soon, Monthly Revenue, and Gross Revenue.
  - Intelligent alert banner highlighting memberships due for expiration within 5 days.
  - Consolidated operational tabs for Quick Overview, Expiring Memberships, and Pending Renewal Requests.
- Built a Dedicated Members Directory (/admin/members):
  - Real-time search engine filtering by member name and email.
  - Multi-dimensional filtering by subscription status (active, expired, suspended, cancelled, pending) and plan tier.
  - Instant status toggle actions (Activate, Suspend, Cancel) executed with immediate database synchronization.
- Built the Add Member Assignment Modal (AddMemberModal.jsx):
  - Allows administrators to assign subscription plans to newly registered users without overwriting admin sessions.
  - Features real-time calculated expiration date previews based on chosen plan durations.

- Built the Offline Payment Recording Desk (RecordPaymentModal.jsx):
  - Facilitates manual payment logging across Cash, UPI / Net Banking, Card Swipe, and Other modes.
  - Automatically sets membership status to active and extends the database expiry date upon payment confirmation.
- Implemented the Renewal & Plan Change Resolution Queue (RenewalRequests.jsx) enabling administrators to resolve pending requests.
- Built a comprehensive Member Detail Investigation View (/admin/members/:id) displaying complete member profile data, current plan metrics, status controls, and full payment audit trails.

Current Status

The MemberPro Membership Management System is a fully operational, end-to-end full-stack web application. All core database tables, security policies, and stored procedures are live on the Supabase PostgreSQL engine. The client interface is fully integrated with live backend data, featuring role-based routing, real-time dashboard analytics, member self-service plan selection, offline payment logging, and automated expiry tracking.

The PostgreSQL schema defines tables for users, plans, memberships, payments, and renewal_requests. The administrative portal provides full control over user lifecycles, payments, and plan oversight, while the member portal enables self-service subscription management. The application has progressed from initial conception into a robust, database-driven software product ready for deployment and evaluation.

Technologies Used

Frontend Framework: React 18 (SPA)
Build Tool & Bundler: Vite
Styling & Design System: Tailwind CSS, PostCSS (Clean Light Theme)
Iconography: Lucide React
Date & Time Utilities: Date-fns
Notifications: React Hot Toast
Routing: React Router DOM (v6)
Database Engine: PostgreSQL (Cloud-hosted via Supabase)
Backend & API Services: Supabase Database API (PostgREST), Supabase Auth (JWT)
Database Language: SQL & PL/pgSQL (Stored Functions, Triggers, RLS Policies)
Development Environment: Visual Studio Code / Windows PowerShell
Database Security: Row-Level Security (RLS) policies, Parametric Stored Procedures (SECURITY DEFINER), and Foreign Key constraints.

Technology Justification:
React delivers a fluid, component-driven user interface with zero-page-reload state updates. Tailwind CSS enables a polished, accessible design system tailored for administrative data density. Supabase provides an enterprise PostgreSQL database eliminating the maintenance overhead of manual backend boilerplate while enforcing strict database-level security through Row-Level Security (RLS) and cryptographic JWT validation.

Project Plan for Future

- Integrate an automated online payment gateway (e.g., Razorpay, Stripe) for automated credit card, debit card, and UPI checkouts.
- Implement automated email and SMS notification triggers (via Supabase Edge Functions / Twilio / Resend) sent 3 days prior to membership expiration.
- Enable automated daily membership expiration jobs using the PostgreSQL pg_cron extension to transition expired accounts without manual intervention.
- Introduce customizable plan creation and pricing modification directly from the admin graphical interface.
- Implement barcode / QR code generation for digital member identification cards and check-in attendance tracking.

- Expand business analytics with graphical charts (e.g., Recharts) illustrating monthly revenue growth, plan distribution, and churn trends.
- Add CSV/PDF export functionality for financial audits, payment receipts, and membership directories.
- Conduct cross-browser, cross-device testing across smartphones, tablets, and desktop workstations.

Expected Outcome

The expected outcome of the MemberPro project is a reliable, secure, and production-ready subscription management platform that bridges administrative operations and customer self-service into a unified ecosystem.

Members gain full visibility into their subscription validity, self-service plan selection, and complete transactional clarity, eliminating ambiguity surrounding payment dates and expiration schedules. Administrators benefit from automated financial calculations, proactive expiration warnings, centralized member record auditing, and swift payment logging.

By centralizing these functions, MemberPro eliminates manual bookkeeping discrepancies, mitigates revenue leakage caused by unmonitored plan expiries, improves member retention, and enforces data integrity across all operational workflows.

Conclusion

The Automated Membership Management and Billing System (MemberPro) demonstrates an efficient application of database design and modern web development principles to solve real-world subscription management challenges. By integrating a reactive frontend with a secure PostgreSQL database, the project unifies user authentication, lifecycle management, offline payment audits, and operational analytics into an intuitive system.

The utilization of React, Tailwind CSS, PostgreSQL, and Supabase establishes a modular, maintainable, and secure software architecture. MemberPro exemplifies how modern database tools, row-level security policies, and stored procedures can automate organizational operations and deliver an exceptional user experience for both administrators and members.

STUDENT **PROJECT GUIDE** **HOD**